Rajeev Jain

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Principal Specialist, Research Software Engineering, Argonne National Laboratory

Professional Experience

Argonne National Laboratory **2009 – Present**
Principal Specialist, Research Software Engineering *Mathematics and Computer Science Division, Chicago, IL*

- SimBoard (E3SM): contributor to the web catalog of E3SM simulation runs, with case metadata, provenance, and links to diagnostics.
- UXarray: lead developer of a Python library for analysis on unstructured climate grids, used at DOE labs and NCAR. Wrote its MCP server, which runs analyses for AI agents on HPC systems.
- E3SM AI platform and ncvis: added the Argo model provider to the E3SM documentation assistant; maintain the SEATS NetCDF viewer.
- SPOTTER-AI (DOE Genesis Mission): attention capture and poisoning experiments on cancer drug-response models.
- SFNO on Aurora: ported a PyTorch climate emulator to Intel GPUs.
- CANDLE and IMPROVE: hyperparameter search workflow (10,000+ runs on Summit, Theta, Cori) and the IMPROVE benchmark framework. R&D 100 Award, 2023.
- FLASH-X: asynchronous checkpoint I/O and compression. R&D 100 Award, 2022.
- MeshKit (PI, DOE NEAMS) and Urban ECP: reactor core mesh generation; coupled weather-to-CFD workflow for downtown Chicago.

University of Chicago **2023 – Present**
Staff At-Large *Consortium for Advanced Science and Engineering (CASE), Chicago, IL*

- Joint appointment for cancer and Earth science software collaborations.

Arizona State University **2007 – 2009**
Research and Teaching Assistant *Structural and Computational Mechanics Lab, Tempe, AZ*

- Finite element research on blast mitigation.

Earlier Experience **2005 – 2007**
Project Engineer, Wipro Technologies | Engineering Intern, Tata Motors *India*

Technical Skills

Languages: Python, C++, Fortran, Bash
Scientific Python: NumPy, Xarray, Numba, PyTorch, Pandas
HPC: MPI, OpenMP, HDF5, NetCDF, PBS and Slurm, Intel and NVIDIA GPUs
Workflow: GitHub Actions, Docker, Parsl, Swift/T, Globus Compute, MCP

Selected Publications

- **Jain, R.** and Jacob, R. "Beyond Tool Execution: Evaluating Scientific MCP Interfaces with UXarray." *AGENT4SC at IEEE eScience*, 2026.
- Partin, A., ..., **Jain, R.**, et al. "Benchmarking community drug response prediction models." *Briefings in Bioinformatics*, 2026.
- **Jain, R.**, Tang, H., Dhruv, A., and Byna, S. "Enabling Data Reduction for Flash-X Simulations." *SC24 / DRBSD-10*, 2024.
- **Jain, R.**, Wozniak, J. M., et al. "Cross-HPO: Optimizing Neural Networks for Cancer Drug Response." *CAFCW at*

SC24, 2024.

- Wozniak, J. M., **Jain, R.**, et al. "CANDLE/Supervisor: A workflow framework for machine learning applied to cancer research." *BMC Bioinformatics*, 2018.

Full list: Google Scholar.

Awards and Service

- R&D 100 Awards: CANDLE (2023), FLASH-X (2022).
- Best Paper, International Meshing Roundtable (2010). ATPESC scholar (2015).
- Program committee, AgenticAI4HPC at SC26 and AGENT4SC at eScience 2026. DOE SBIR/STTR reviewer.

Education

University of Chicago 2020
M.S. in Computer Science

Arizona State University 2009
M.S. in Structural Engineering Graduate Fellowship Recipient

Indian Institute of Technology (ISM) Dhanbad 2006
B.Tech. in Mechanical Engineering